01 Getting started with selenium

In [1]:
```
from selenium import webdriver
import time
```

In [4]:
```
driver = webdriver.Chrome()
driver.maximize_window()

url = "https://www.google.com"
driver.get(url)
time.sleep(2)

print(f"Title: {driver.title}")
print(f"Url: {driver.current_url}")

driver.save_screenshot("google-screen.png")
print("screenshot taken successfully")

driver.quit()
print("browser closed successfully")
```

```
Title: Google
Url: https://www.google.com/
screenshot taken successfully
browser closed successfully
```

02 Navigate webpage

In [1]:
```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.keys import Keys
```

```
    1. Typing input into field
    2. Clearing input field
```

In [13]:
```
driver = webdriver.Firefox()
driver.maximize_window()

url = "https://www.google.com"
driver.get(url)
time.sleep(2)

search_bar = driver.find_element(by = By.XPATH, value = '//*[@id="APjFqb"]')
search_bar.send_keys("machine learnengs")
time.sleep(2)

search_bar.clear()
time.sleep(2)

search_bar.send_keys("Machine Learning")
time.sleep(2)
```

```
search_bar.send_keys(Keys.ENTER)
time.sleep(2)

driver.quit()
```

```
    1. Typing input into field
    2. Clearing input field
    3. Clicking the button
    4. Submitting the form
```

In [22]:
```
driver = webdriver.Chrome()
driver.maximize_window()

url = "https://github.com/login"
driver.get(url)
time.sleep(2)

username_field = driver.find_element(by = By.ID, value = 'login_field')
username_field.send_keys("abc")
time.sleep(2)
username_field.clear()
time.sleep(2)
username_field.send_keys("your username")

password_field = driver.find_element(by = By.ID, value = 'password')
password_field.send_keys("abc")
time.sleep(2)
password_field.clear()
time.sleep(2)
password_field.send_keys("your password")

login_button = driver.find_element(by = By.XPATH, value = "//input[@name='commit
login_button.click()
time.sleep(2)
```

Working on dropdown

In [16]:
```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.select import Select
```

In [36]:
```
driver = webdriver.Chrome()
driver.maximize_window()

url = "https://www.miniclip.com/careers/vacancies"
driver.get(url)
time.sleep(2)

locations_field = driver.find_element(by = By.XPATH, value = '//fieldset[2]//sel
# driver.execute_script("window.scrollBy(0, 250);")
location_dropdown = Select(locations_field)
time.sleep(3)

location_dropdown.select_by_index(1)
time.sleep(3)
```

```
location_dropdown.select_by_index(0)
time.sleep(3)

location_dropdown.select_by_visible_text("London, United Kingdom")
time.sleep(3)

driver.quit()
```

Working on multiselect

In [33]:
```
driver = webdriver.Chrome()
driver.maximize_window()
time.sleep(2)

url = "https://demoqa.com/select-menu"
driver.get(url)
time.sleep(2)

cars_element = driver.find_element(by = By.XPATH, value = '//*[@id="cars"]')
# driver.execute_script("arguments[0].scrollIntoView();", cars_element)
cars_ms = Select(cars_element)
time.sleep(2)

cars_ms.select_by_index(1)
time.sleep(2)

cars_ms.select_by_visible_text("Opel")
time.sleep(2)

cars_ms.select_by_visible_text("Audi")
time.sleep(2)

cars_ms.deselect_by_index(1)
time.sleep(2)

cars_ms.deselect_by_visible_text("Audi")
time.sleep(2)

cars_ms.deselect_all()
time.sleep(2)

driver.quit()
```

Working on scrolling

In [29]:
```
driver = webdriver.Firefox()
driver.maximize_window()

url = "https://en.wikipedia.org/wiki/Machine_learning"
driver.get(url)
time.sleep(2)

# scrolling to specific element
ai_subtopic = driver.find_element(by = By.XPATH, value = '//*[@id="Artificial_in
driver.execute_script("arguments[0].scrollIntoView();", ai_subtopic)
```

```
time.sleep(2)

# scrolling vertically
driver.execute_script("window.scrollBy(0, 1000);")
time.sleep(2)
driver.execute_script("window.scrollBy(0, -1000);")
```

In [32]:

```
driver = webdriver.Firefox()
driver.maximize_window()

url = "https://en.wikipedia.org/wiki/Machine_learning"
driver.get(url)
time.sleep(2)

# # scrolling to specific element
# ai_subtopic = driver.find_element(by = By.XPATH, value = '//*[@id="Artificial_
# driver.execute_script("arguments[0].scrollIntoView();", ai_subtopic)
# time.sleep(2)

# # scrolling vertically
# driver.execute_script("window.scrollBy(0, 1000);")
# time.sleep(2)
# driver.execute_script("window.scrollBy(0, -1000);")

# scrolling page height
driver.execute_script("window.scrollTo(0, document.body.scrollHeight);")
time.sleep(2)
driver.execute_script("window.scrollTo(0, -document.body.scrollHeight);")
```

In [2]:

```
# Infinite scrolling

driver = webdriver.Firefox()
driver.maximize_window()

url = """
https://www.google.com/search?sca_esv=28176cee0d949075&sxsrf=ANbL-n4Wm1KRfLb2PKb
&udm=2&fbs=ADc_l-aN0CWEZBOHjofHoaMMDiKpaEWjvZ2Py1XXV8d8KvlI3ppPEReeCOS7s1VbbZz2T
&q=csk&sa=X&ved=2ahUKEwio7dPv7K2TAxWZzDgGHbVVAtQQtKgLegQIDhAB&biw=1536&bih=730&d
"""
driver.get(url)
time.sleep(2)

prevh = driver.execute_script("return document.body.scrollHeight")

while True:
    print(f"Webpage height: {prev_height:,} pixels")
    driver.execute_script("window.scrollTo(0, document.body.scrollHeight);")
    time.sleep(5)
    curh = driver.execute_script("return document.body.scrollHeight")
    if prevh == curh:
        break

    prevh = curh
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[2], line 17
     14 prevh = driver.execute_script("return document.body.scrollHeight")
     16 while True:
---> 17     print(f"Webpage height: {prev_height:,} pixels")
     18     driver.execute_script("window.scrollTo(0, document.body.scrollHeigh
t);")
     19     time.sleep(5)

NameError: name 'prev_height' is not defined
```

03 Advanced Web Interactions

In [1]:
```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.keys import Keys
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
```

In [5]:
```
driver = webdriver.Chrome()
driver.maximize_window()

url = "https://www.google.com"
driver.get(url)
time.sleep(2)

search_bar = driver.find_element(by = By.ID, value = "APjFqb")
search_bar.send_keys("Machine Learning")

wait = WebDriverWait(driver, 5)
wait.until(EC.element_to_be_clickable((By.XPATH, "//div[@class='FPdoLc lJ9FBc']/

search_bar.send_keys(Keys.ENTER)
```

In [6]:
```
driver = webdriver.Chrome()
driver.maximize_window()

driver.implicitly_wait(5)

url = "https://www.google.com"
driver.get(url)
time.sleep(2)

search_bar = driver.find_element(by = By.ID, value = "APjFqb")
search_bar.send_keys("Machine Learning")
time.sleep(2)

search_bar.send_keys(Keys.ENTER)
time.sleep(2)

driver.quit()
```

Working on IFrames

In [13]:
```
driver = webdriver.Chrome()
driver.maximize_window()

url = "https://www.w3schools.com/html/tryit.asp?filename=tryhtml_iframe_target"
driver.get(url)
time.sleep(2)

# switch context to frame
iframe_element = driver.find_element(by = By.XPATH, value = '//*[@id="iframeResu
driver.switch_to.frame(iframe_element)
time.sleep(2)

link = driver.find_element(by = By.XPATH, value = '/html/body/p[1]/a')
link.click()
time.sleep(2)

# switch back to main content
driver.switch_to.default_content()
driver.quit()
```

Handling Alerts

Javascript alerts

In [8]:
```
driver = webdriver.Chrome()
driver.maximize_window()

url = "https://www.w3schools.com/js/tryit.asp?filename=tryjs_alert"
driver.get(url)
time.sleep(2)

iframe_element = driver.find_element(By.XPATH, '//*[@id="iframeResult"]')
driver.switch_to.frame(iframe_element)

button = driver.find_element(By.XPATH, '/html/body/button')
button.click()
time.sleep(2)

print(f"Alert: {driver.switch_to.alert.text}")
driver.switch_to.alert.accept()
time.sleep(2)

driver.switch_to.default_content()
driver.quit()
```

```
Alert: I am an alert box!
```

Confirmation alerts

In [10]:
```
driver = webdriver.Chrome()
driver.maximize_window()

url = "https://www.w3schools.com/js/tryit.asp?filename=tryjs_confirm"
driver.get(url)
time.sleep(2)
```

```
iframe_element = driver.find_element(by = By.ID, value = 'iframeResult')
driver.switch_to.frame(iframe_element)

button = driver.find_element(by = By.XPATH, value = '/html/body/button')
button.click()
time.sleep(2)

print(f"Alert: {driver.switch_to.alert.text}")
driver.switch_to.alert.accept()
print("You clicked accept")
time.sleep(5)

button = driver.find_element(by = By.XPATH, value = '/html/body/button')
button.click()
time.sleep(2)

print(f"Alert: {driver.switch_to.alert.text}")
driver.switch_to.alert.dismiss()
print("You clicked cancel")
time.sleep(5)

driver.switch_to.default_content()
driver.quit()
```

```
Alert: Press a button!
You clicked accept
Alert: Press a button!
You clicked cancel
```

Prompt alerts

In [15]:

```
driver = webdriver.Chrome()
driver.maximize_window()

url = "https://www.w3schools.com/js/tryit.asp?filename=tryjs_prompt"
driver.get(url)
time.sleep(2)

iframe_element = driver.find_element(by = By.ID, value = 'iframeResult')
driver.switch_to.frame(iframe_element)

button = driver.find_element(by = By.XPATH, value = '/html/body/button')
button.click()
time.sleep(2)

print(f"Alert: {driver.switch_to.alert.text}")
driver.switch_to.alert.send_keys("Are you ready???")
time.sleep(2)

driver.switch_to.alert.accept()
print("You clicked OK")
time.sleep(2)

driver.switch_to.default_content()
driver.quit()
```

```
Alert: Please enter your name:
You clicked OK
```

In [21]:
```
driver = webdriver.Chrome()
driver.maximize_window()

url = "https://www.w3schools.com/js/tryit.asp?filename=tryjs_prompt"
driver.get(url)
time.sleep(2)

iframe_element = driver.find_element(by = By.ID, value = 'iframeResult')
driver.switch_to.frame(iframe_element)

button = driver.find_element(by = By.XPATH, value = '/html/body/button')
button.click()
time.sleep(2)

print(f"Alert: {driver.switch_to.alert.text}")
driver.switch_to.alert.send_keys("Are you ready???")
time.sleep(2)

driver.switch_to.alert.accept()
print("You clicked OK")
time.sleep(2)

button = driver.find_element(by = By.XPATH, value = '/html/body/button')
button.click()
time.sleep(2)

driver.switch_to.alert.dismiss()
print("You clicked cancel")
time.sleep(2)

driver.switch_to.default_content()
driver.quit()
```

```
Alert: Please enter your name:
You clicked OK
You clicked cancel
```

Woking on Page Object Model(POM)

In [17]:
```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By
```

In [20]:
```
driver = webdriver.Chrome()
driver.maximize_window()

url = "https://github.com/login"
driver.get(url)
time.sleep(2)

driver.implicitly_wait(5)

class Login:
    def __init__(self, driver):
        self.driver = driver
        self.username = (By.ID, "login_field")
        self.password = (By.ID, "password")
        self.login_button = (By.XPATH, "//input[@name='commit']")
```

```python
    def login(self, username, password):
        self.driver.find_element(*self.username).send_keys(username)
        self.driver.find_element(*self.password).send_keys(password)
        time.sleep(2)
        self.driver.find_element(*self.login_button).click()


login_page = Login(driver)
login_page.login("your username", "your password")
time.sleep(2)

driver.quit()
```

In [ ]: